

## Curriculum Vitae

# Diogo Afonso Pedro Nunes

(alias, Diogo A.P. Nunes)



NLP & Machine Learning Researcher @ INESC-ID/HLT  
PhD Candidate (defense pending 2025) @ IST  
Lisbon, Portugal

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[Google Scholar](#) | [LinkedIn](#) | [Website](#)

## SUMMARY

Natural Language Processing (NLP) researcher with 5+ years of experience applying machine learning to real-world health and language challenges. My recent work blends large language models (LLMs), clinical and patient language data, and patient-centered design to empower patients and support medical decision-making. I've collaborated nationally and internationally with academic hospitals and research labs (Portugal, US) and recently led a team to *1st place* at eRisk 2025 for detecting depression symptoms in text. Currently completing a PhD in Computer Science, focused on applying NLP to chronic pain assessment and patient support.

## SKILLS & TECHNOLOGIES

**ML/NLP Tools:** PyTorch, Hugging Face Transformers, scikit-learn, TensorFlow, vLLM, Git  
**Techniques:**

- LLM fine-tuning & continual pretraining
- Few-shot prompting, RAG (retrieval-augmented generation)
- Data synthesis with LLMs
- Language model interpretability
- Clinical NLP, clinical state prediction
- Other: web scraping (80K+ entries), collaborative research and data collection protocols

## EXPERIENCE

**Researcher** *INESC-ID, HLT lab | Lisbon, Portugal. May 2019 – Present*

Designed NLP systems for healthcare, including tools to interpret patient pain narratives and predict clinical states. Led LLM-focused projects supporting patient empowerment and self-expression in chronic pain contexts. Collaborated with São João Hospital (Porto, Portugal) and Northwestern University (Chicago, US) on joint research initiatives with chronic pain patients. Co-authored award-winning and peer-reviewed studies. *1st place @ eRisk 2025* (depression symptom detection in text, CLEF competition). Applied for and won two research grants.

**IT Infrastructure Solutions** *INESC-ID | Lisbon, Portugal. Jan 2022 – Dec 2024*

Designed and developed internal intranet and IT solutions for researcher and staff database, Technologies: .NET, docker, Git

**Teaching Assistant** *Instituto Superior Técnico (IST) | Lisbon, Portugal. Feb 2020 – Jul 2020*

**Internships (Software & Data Science)** *EDP Innovation, CEiiA, Building Global Innovators | Lisbon/Porto, Portugal. 2017 – 2019*

## INVITED TALKS & LECTURES (selected)

**“Computational Models of the Language of Pain: Exploring Patient Narratives with Natural Language Processing”**

Keynote Speaker @ Aarhus University, Sep 2025

**“How to Save a Life: Redefining Healthcare with AI-Wearables”**

Teacher, BEST Summer Course @ IST, Jul 2025

**“Computational assessment of chronic pain based on patient language of pain”**

Guest Lecture, ISCTE, Computational Language Processing, Jun 2025

**“Every Step Matters: Stories from the PhD Journey”**

Panelist, PhD Open Days @ IST, Nov 2024

## EDUCATION

**PhD in Computer Science** (defense pending 2025)

*Instituto Superior Técnico (IST) | Lisbon, Portugal*

Feb 2021 – Sep 2025

Research: NLP for healthcare. Collaboration with FMUP (Porto, Portugal; co-supervision). Research rotation and collaboration with Northwestern University (Chicago, US). Thesis submitted May 2025.

**MSc + BSc in Computer Science**

*Instituto Superior Técnico (IST) | Lisbon, Portugal*

Sep 2015 – Jan 2021

ERASMUS exchange at EFREI, Paris, France

## AWARDS & RECOGNITION (selected)

- Best Clinical Research Paper – Portuguese Association for the Study of Pain
- 1st Place, eRisk 2025 challenge – Depression symptom detection
- Finalist, “3 Minute Thesis” Competition (2023), University of Lisbon

## PUBLICATIONS (selected)

*(Full list available on Google Scholar)*

Nunes et al., *INESC-ID @ eRisk 2025: Exploring Fine-Tuned, Similarity-Based, and Prompt-Based Approaches to Depression Symptom Identification*. Accepted @ CLEF Sep 2025, Madrid. Preprint: [10.48550/arXiv.2506.02924](https://arxiv.org/abs/10.48550/arXiv.2506.02924)

Nunes et al., *Advancing the prediction and understanding of placebo responses in chronic back pain using large language models*. Under review. 2025. Preprint: [10.1101/2025.01.21.25320888](https://arxiv.org/abs/10.1101/2025.01.21.25320888)

Nunes et al., *Modeling chronic pain experiences from online reports using the Reddit Reports of Chronic Pain dataset*. Information, MDPI. 2023. DOI: [10.3390/info14040237](https://doi.org/10.3390/info14040237)